



CHAPTER 6

Fundamental Concepts of Macroeconomics

Worku A.

Introduction

- ❑ Macroeconomics, studies about overall or aggregate behavior of the economy, such as:
 - Economic growth,
 - Employment,
 - Inflation,
 - Distribution of income,
 - International trade.
 - How budgetary deficit and trade deficit occur and
 - Macroeconomic policies

Goals of Macroeconomics

❑ Macroeconomics it aimed at how;

- To achieve high economic growth
- To reduce unemployment
- To attain stable prices
- To reduce budget deficit and balance of payment (BoP) deficit
- To ensure fair distribution of income

The National Income Accounting

- ❑ **National Income Accounting (NIA)** is an accounting record of the level of economic activities of an economy.

- ❑ ***It is important in many ways as;***
 - It enables us to measure the level of total output in a given period of time, and to explain the causes for such level of performance.
 - It enables us to observe the long run trend of the economy.
 - It provides information to formulate policies and design plans.

Gross Domestic Product (GDP)

- ❑ It is the total value of currently produced final goods and services that are produced ***within a country's boundary*** during a given period of time, usually one year, irrespective of who produce it.
- ❑ Thus, we can note that GDP measures only;
 - Currently produced goods & services
 - Final goods and services
 - The value of ***domestically*** produced g & s. and
 - It takes the market value of goods and services.

$$GDP = \sum P_i Q_i$$

Where, Q_i and P_i are series of final g&s and their corresponding prices respectively.

Market value

- To compute the total value of different goods and services, the national income accounts use market prices.

- Thus, if



$$\begin{aligned}\text{GDP} &= (\text{Price of apples} \times \text{Quantity of apples}) \\ &\quad + (\text{Price of oranges} \times \text{Quantity of oranges}) \\ &= (\$0.50 \times 4) + (\$1.00 \times 3) \\ \text{GDP} &= \$5.00\end{aligned}$$

- ***In short:***

$$\begin{aligned}\text{GDP} &= (\text{price of apples} * \text{quantity of apples}) \\ &+ (\text{price of oranges} * \text{quantity of oranges})\end{aligned}$$

Gross National Product (GNP)

- It is the total value of final goods and services currently produced by ***domestically owned factors of production*** in a given period of time, usually one year, irrespective of their geographical locations.

$$\text{GNP} = \text{GDP} + \text{NFI}$$

- Net factor income (**NFI**) = factor payments from abroad – factor payments to abroad.

If $\text{NFI} > 0$, then $\text{GDP} < \text{GNP}$

If $\text{NFI} < 0$, then $\text{GDP} > \text{GNP}$

If $\text{NFI} = 0$, then $\text{GDP} = \text{GNP}$

Approaches to measure national income (GDP/GNP)

□ There are three approaches to measure GDP/GNP. These are;

- I. Product/value added approach,
- II. Expenditure approach and
- III. Income approach

I. Product Approach:

- ❑ In this approach, GDP is calculated by adding the market value of goods and services currently produced by each sector of the economy.
- ❑ In this case, GDP includes only the values of ***final goods and services*** in order to avoid double counting.
- ❑ Example, the components of a car that are sold to the manufacturers are called intermediate goods, and their value is not included in GDP.

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- ❑ There are two possible ways of avoiding double counting.
 - Taking only the value of final goods and services
 - Taking the sum of the valued added by all firms at each stage of production
- ❑ Let's have the following hypothetical examples:

Taking the value of final goods & services

Sectors	Value of Output (in million birr)
Agriculture and allied activities	9309
- Agriculture	7000
- Forestry	1000
- Fishing	1309
Industry	147413
- Mining & quarrying	9842
- Large & medium scale manufacturing	91852
- Electricity & water	13717
- Construction	32002
Service	357 872
- Banking insurance and real estate	121704
- Public administration & defense	36605
- Health	20000
- Education	32509
- Domestic & other services	147054
4. Net factor income from abroad	<u>87348</u>

$$\text{GDP} = 9,309 + 147,413 + 357,872 = \underline{\underline{514,594}}$$

$$\text{GNP} = \text{GDP} + \text{NFI} = 514,594 + 87,348 = \underline{\underline{601,942}}$$

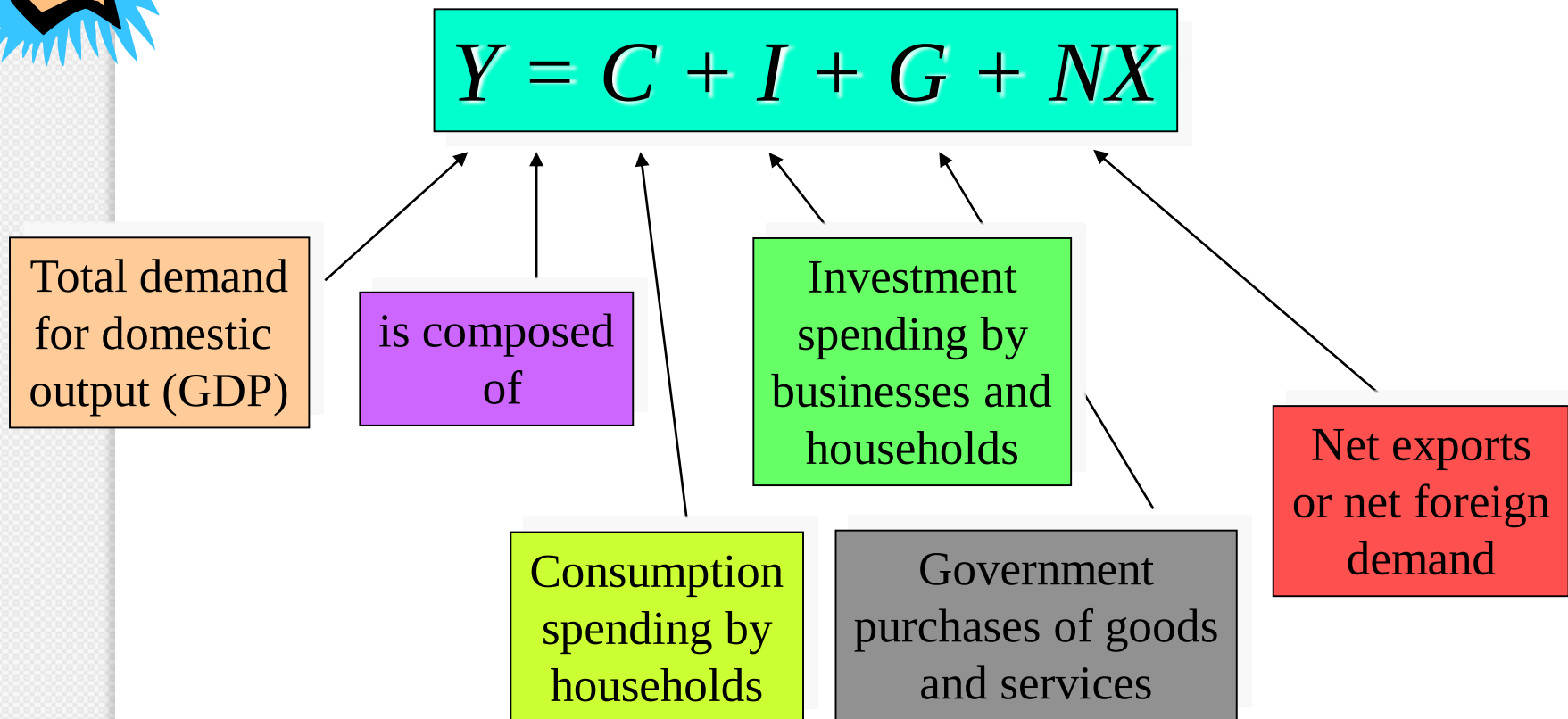
Taking the sum of the valued added by all firms at each stage of production

Stages of production	Values of output (in birr)	Cost of intermediate inputs	Value added
Farmer	500	0	500
Oil factory	2000	500	1500
Retailers	2500	2000	500

- ❑ If all in column 2 were added GDP would be $5000 = (500 + 2000 + 2500)$.
- ❑ But there is problem of double counting which equals 2500, because of considering intermediate input in the calculation.
- ❑ But ***GDP*** = **$500 + 1500 + 500$**



2. Expenditure Approach.



This is the called the *national income accounts identity*.

.... Expenditure Approach:

- ❑ In this approach GDP is measured by adding all expenditures on final goods and services produced in the country by all sectors of the economy.

$$\Rightarrow \mathbf{GDP = C + I + G + NX}$$

1. Personal consumption expenditures (**C**): household spending on consumer goods.

- ❑ It includes expenditures by households on:
 - ✓ Durable consumer goods; automobiles, refrigerators, video recorders, etc.,
 - ✓ Non-durable consumer goods (clothes, shoes, pens, food etc.) and services.

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2. Gross private domestic investment (*I*):

□ *Main categories*

- ✓ ***Residential investment***: spending of households on the construction of new houses,
- ✓ ***Business fixed investment***: spending of firms on buildings and equipment for business use.
- ✓ ***Inventory investment***: the change in inventories of firms

**Net private domestic investment = gross private domestic investment
- depreciation.**

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3. *Government consumption (G):*

- ☐ Includes spending on;
 - ✓ Finished products
 - ✓ Direct purchases of resources

- ☐ Excludes government transfer payments as they do not reflect current production although they are part of government expenditure.

4. Net exports (NX)

$$\Rightarrow NX = EX - IM$$

Where, EX = Exports and IM = imports values of g&s.¹⁶

Cont'd

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Hypothetical example

Types of expenditure	Amount (in million Birr)
1. Personal consumption expenditure	4500
Durable consumer goods	1500
Non-durable consumer goods	2000
Services	1000
2. Gross private domestic investment	600
Business fixed investment	250
Construction Expenditure	300
Increases in inventories	50
3. Government expenditure on goods and services	250
Federal government	100
State government	150
4. Net export	-50
Exports	150
Imports	200
GDP at current market price	5300